BABY

Build A Better Yield blanket

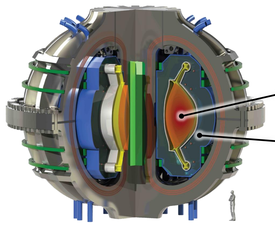
Nikola Goleš, Rémi Delaporte-Mathurin, Andrew T. Lanzrath, Stefano Segantin, Collin S. Dunn, Emily Edwards, Weiyue Zhou, Rui Vieira, Lihua Zhou, Rick Leccacorvi, Sara E. Ferry, Ethan E. Peterson, Dennis G. Whyte, Kevin B. Woller
✉kbwoller@mit.edu

MIT Plasma Science and Fusion Center

arpa·e OPEN 2021

Tritium breeding

2030



$$D + T \rightarrow He + n$$

$$Li + n \rightarrow T + He$$

The ARC concept relies on the Liquid Immersion Blanket for breeding tritium

$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Can we test this concept now?

Goals of BABY and LIBRA

☑☐ To demonstrate the concept of Liquid Immersion Blanket

To improve the Technology Readiness Level of LIB

To validate numerical models for tritium transport

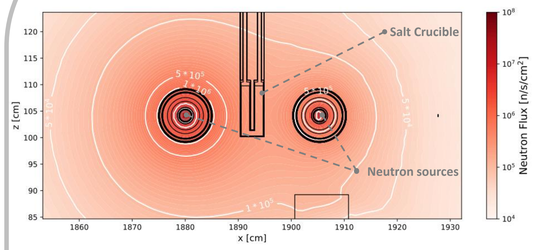
To gain knowledge on molten salt behavior in fusion environments

Neutrons

Neutronics

- Neutronics simulations track the path of neutrons
- Used to predict tritium generation, neutron flux, heating, production of activated materials…

Neutron flux
$n/cm^2/s$



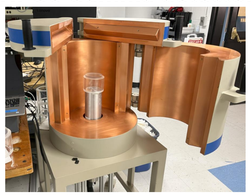
Detection

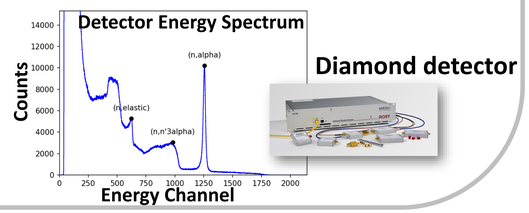
- LIBRA will provide neutronics data for validating codes
- Activation foils and diamond detectors as used to characterize the neutron sources and track the neutron flux

Activation foils

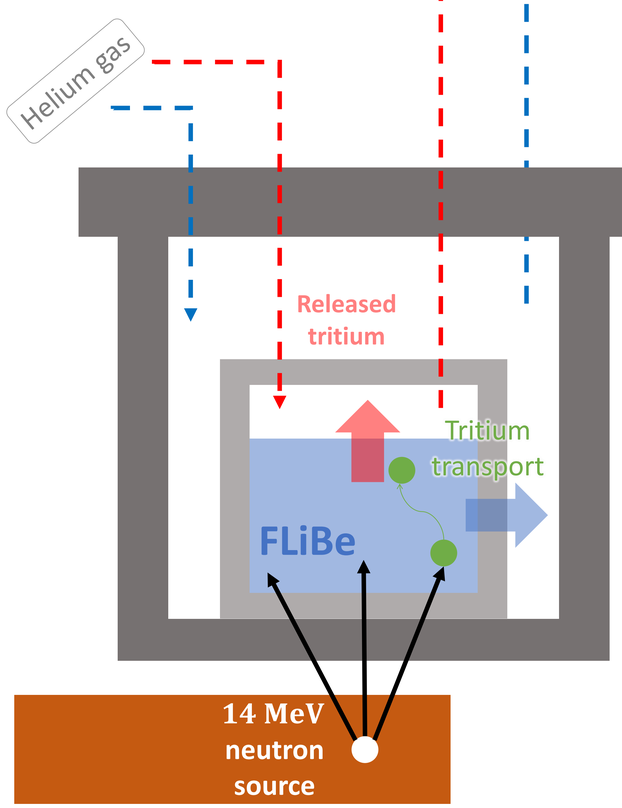


HPGe Gamma detector





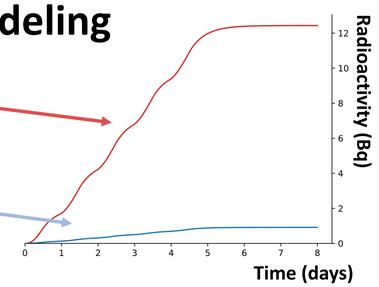
Diamond detector



Permeated tritium

Tritium

Modeling



0D tritium transport model:

$$V\frac{\partial c}{\partial t} = S - A_p\ k_p\ c - A_r\ k_r\ c$$

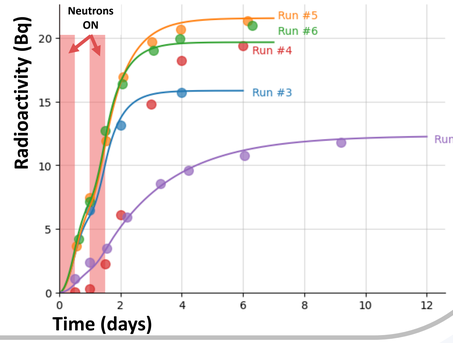
V: salt volume
c: salt tritium concentration
S: tritium source
A_i: surface area
k_i: mass transport coefficient

Detection

Tritium water bubblers

- Bubblers collect 99% of the tritium
- Water samples are analyzed with Liquid Scintillation Counting
- Measurements on BABY serve as validation data

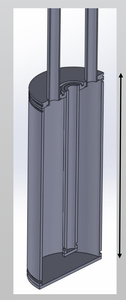
0.1 mL BABY Top release measurements



BABY specs

Salt volume	$0.1 - 1$ L
TBR	$\approx 10^{-3}$
Temperature	700 °C
DT Neutron energy	14 MeV

CAD Models



100 mL

